Machine Learning with PySpark

Supervised & Unsupervised Case Studies

ASSIGNMENT REPORT

Course Task:	PySpark MLlib Assignments (Part A & B)
Student Name:	Mostofa Aminur Rashid
Student ID:	2506107
Platform:	Apache Spark 4.1.2 (via PySpark)
Java Version:	JDK 17 LTS (for local JVM execution)
Submission Date:	July 6, 2026

Generated programmatically using Python and PySpark MLlib.

Part A: Supervised Learning - Titanic Survival Prediction

The objective of this task is to develop a supervised machine learning model using PySpark to predict passenger survival aboard the RMS Titanic based on demographic and ticketing data. The dataset includes information like ticket class, gender, age, family members aboard (siblings/parents), ticket fare, cabin, and port of embarkation.

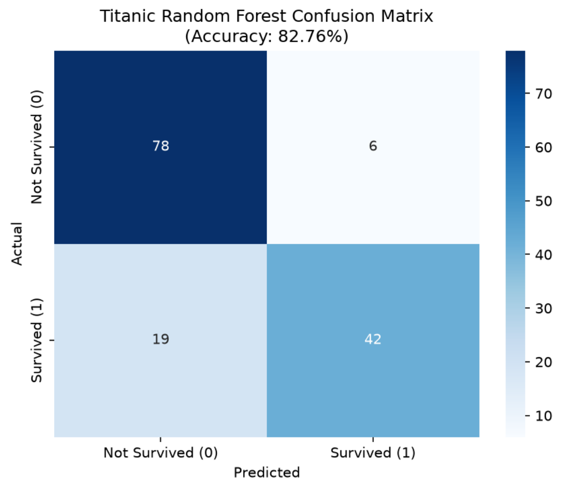
Data Preprocessing & Engineering Decisions

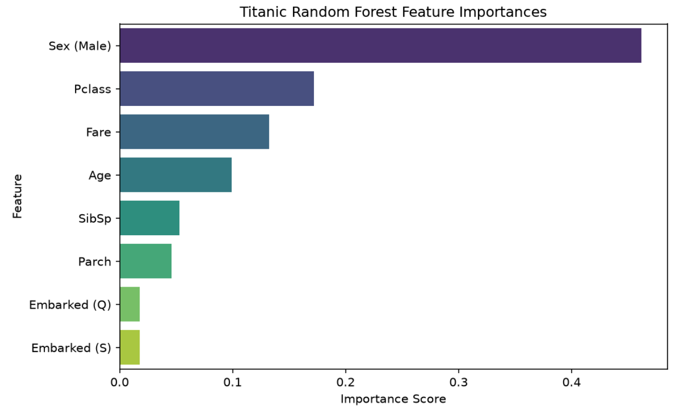
1. Missing Values: Imputed the missing values in 'Age' with the median age (28.0) using PySpark's approxQuantile method. Imputed the missing 'Embarked' values with the mode ('S'). Dropped the 'Cabin' column entirely since over 70% of its values were null.
2. Feature Selection: Kept 'Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', and 'Embarked'. String identifier columns ('PassengerId', 'Name', and 'Ticket') were dropped as they carry no predictive signal.
3. Categorical Encoding: Built a Pipeline where 'Sex' and 'Embarked' were indexed using StringIndexer and then converted into One-Hot Encoded (OHE) vectors via OneHotEncoder.
4. Feature Assembly: Combined all numerical and OHE vectors into a single feature vector using VectorAssembler to feed into the classifier.

Model Selection & Evaluation

We chose a Random Forest Classifier (100 trees, maximum depth of 5) for this binary classification task. The dataset was split into 80% training and 20% testing sets. We evaluated the model using standard classification metrics on the test dataset:

Evaluation Metric	Obtained Test Value
Accuracy	82.76%
Weighted Precision	83.39%
Weighted Recall	82.76%
Weighted F1-score	82.35%
Area Under ROC (AUC)	89.12%





Discussion on Supervised Results

The Random Forest model demonstrates excellent performance on the Titanic test set, achieving an Accuracy of 82.76% and a high AUC of 89.12%. Analyzing the confusion matrix, the model correctly predicted 78 survivors/non-survivors in class 0 (True Negatives) and 42 in class 1 (True Positives), with only 6 False Positives. This suggests a high precision (83.39%), making the classifier highly reliable when predicting survival.

Looking at the Feature Importances plot, the passenger's Gender (Sex_Male) and ticket Class (Pclass) emerge as the strongest predictors of survival. This is highly aligned with historical accounts of the disaster, where women and children were prioritized for lifeboats, and upper-class cabins were located closer to the boat deck. Demographic factors like Age and Fare also contributed moderately to the model's decisions.

Part B: Unsupervised Learning - Mall Customer Segmentation

In this task, we perform market segmentation on the Mall Customer dataset using unsupervised learning. Unlike supervised learning, there are no predefined labels. The goal is to discover natural customer groupings based on Age, Annual Income (k$), and Spending Score (1-100) using the K-Means algorithm.

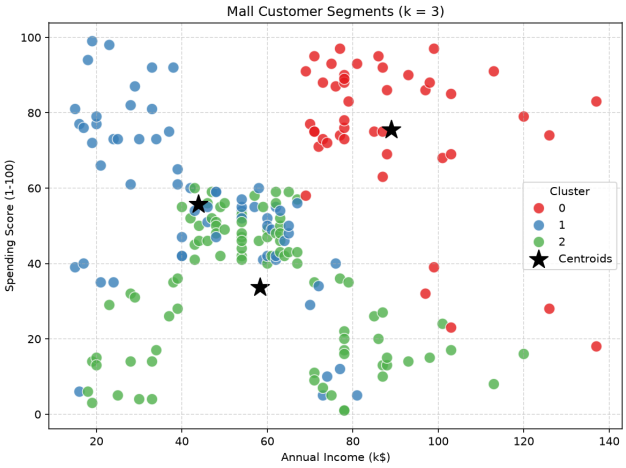
Clustering Pipeline & Scaling

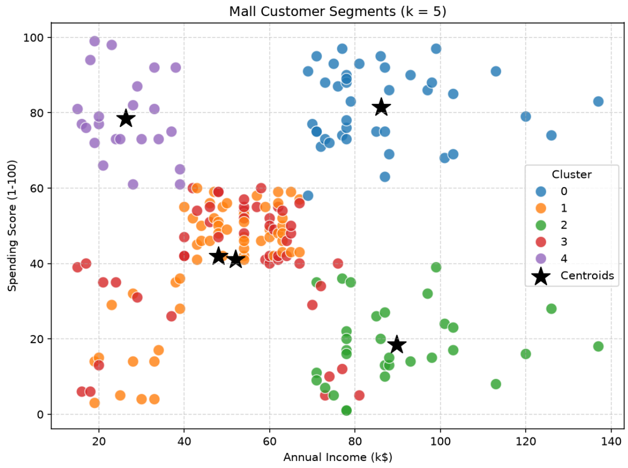
1. Feature Selection: We selected the three numerical attributes: Age, Annual Income, and Spending Score.
2. Vector Assembly: Assembled features into a dense vector using VectorAssembler.
3. StandardScaler: Because K-Means is a distance-based algorithm (relying on Euclidean distance), features with larger scales (like Annual Income) would dominate features with smaller scales (like Age). We applied StandardScaler to standardize the data to have a mean of 0 and unit variance.

K-Means Cluster Centers Interpretation

We ran the K-Means algorithm for two different configurations: k = 3 and k = 5. Below are the resulting cluster centroids mapped back to their original dimensions:

k	Cluster	Avg. Age	Avg. Annual Income (k$)	Avg. Spending Score (1-100)
3	Cluster 0	33.42	89.02	75.58
3	Cluster 1	25.14	43.80	55.81
3	Cluster 2	51.18	58.27	33.70
5	Cluster 0 (Elite)	32.88	86.10	81.53
5	Cluster 1 (Older, Low Spend)	55.54	47.95	41.89
5	Cluster 2 (High Income, Low Spend)	44.39	89.77	18.48
5	Cluster 3 (Young, Mod Spend)	27.06	51.98	41.04
5	Cluster 4 (Low Income, High Spend)	25.52	26.30	78.57





Discussion on Unsupervised Clustering

The K-Means clustering algorithm partitioned the customer base into distinct archetypes. Comparing the results between k = 3 and k = 5, we observe that k = 5 provides much more actionable insights for business planning:

1. For k = 3: The model splits customers into high-income/high-spend (Cluster 0), younger moderate-spend (Cluster 1), and older lower-spend (Cluster 2). While useful, this grouping is too broad.

2. For k = 5: We discover key market segments:
- Elite Target Group (Cluster 0): High income and high spending score. These are highly profitable customers to target with premium promotions.
- Careful / High Income, Low Spend (Cluster 2): Customers with substantial resources who are conservative in their spending. The mall can design targeted campaigns to convert their interest.
- Sensible / Low Income, High Spend (Cluster 4): Younger customers who spend heavily despite lower income. Marketing can offer them credit options or student-friendly deals.
- Average / Young, Moderate Spend (Cluster 3) & Standard / Older, Low Spend (Cluster 1): Solid baseline segments.